

λ **HAOBIN (HIROKI) CHEN** λ

+1 (812) 325-6706 $\diamond$ hiroki.chen.us@gmail.com $\diamond$ <https://hiroki-chen.github.io>

EDUCATION

Indiana University Bloomington, IN, USA

2023 - Dec. 2027 (*Expected*)

Ph.D. in Computer Science

Advisors: Haixu Tang, Chenghong Wang (IU) & XiaoFeng Wang (Nanyang Technological University)

Nankai University, Tianjin, China

2019 - 2023

B.Eng. in Information Security

RESEARCH INTERESTS

Computer Security & Privacy; Confidential Computing (TEEs); Formal Verification & PL (Verus, Coq); Systems Security; Privacy-Enhancing Technologies.

INDUSTRIAL EXPERIENCE

Ant Group

May 2026 - Sept. 2026

Research Intern

Beijing, China

- Continue formal verification of the **Asterinas** kernel, focusing on high-level security property proofs.

CertiK Inc.

Oct. 2025 - Mar. 2026

Security Research Intern (Mentors: Dr. Hao Chen, Dr. Sean N. Anderson)

New York, NY

- Applied **Verus** to verify the memory management module of the **Asterinas** Rust-based OS kernel (ATC'25), ensuring spatial and temporal memory safety.
- Conducted formal verification of Solana smart contracts and protocols to provide mathematical assurance for digital assets.

Privacy Innovation Lab, TikTok Inc.

May 2024 - Aug. 2024

Research Intern (Mentor: Dr. Mingshen Sun)

San Jose, CA

- Proposed **TAVERNS**, a decentralized remote attestation paradigm that eliminates reliance on centralized verification services.
- Developed a TEE-based verification solution using **Zero-Knowledge Proofs (ZKP)**, resulting in a **U.S. Patent application**.
- Engineered data clean rooms (ManaTEE, part of the *Confidential Computing Consortium*) for TikTok's research platform on Google Cloud TEE instances using k8s.

Google Summer of Code: Apache Teaclave

Jun. 2023 - Nov. 2023

Open Source Contributor

Remote

- Developed data analysis solutions for privacy policy enforcement using Trusted Execution Environments.
- Integrated formal verification components to enhance the framework's security guarantees.

ACADEMIC EXPERIENCE

Verifiable Policy Enforcement in Confidential VMs

Jul. 2026 - Present

Research Assistant (with Prof. Chenghong Wang, Mingshen Sun & Benjamin S. Ternthal)

IU

- Designed intra-CVM isolation to protect trusted governance logic from an untrusted guest OS.
- Built a Verus-verified IFC monitor for policy enforcement and provenance tracking.
- Bound provenance and policy events to hardware attestation for cloud-native workloads.

Privacy-Preserving Data Sharing and Analytics (PPDSA) Aug. 2024 - Present
Research Assistant (with Prof. XiaoFeng Wang, Chenghong Wang (IU), Marcela Milara (Intel)) IU

- Applied state-of-the-art confidential computing technology to securing data analytics at large scale.
- Applied rigorous machine-assisted verification and policy enforcement for data lakehouses.
- Explored hardware-software co-designs for modern confidential data analytics.

Center for Distributed Confidential Computing (CDCC) Aug. 2023 - Present
Research Assistant (with Prof. Haixu Tang & XiaoFeng Wang) IU

- Architected hardware-software co-designs for policy-enforced isolation within TEEs.
- Utilized **Verus** and **Coq** to verify large-scale codebases and security-critical system kernels.
- Optimized secure system performance for CPU/GPU TEEs in cloud and edge environments.

Proof of Being Forgotten: Rust-SGX Enclave Framework May 2022 - Jun. 2023
Research Assistant (with Prof. XiaoFeng Wang & Dr. Mingshen Sun) Remote

- Implemented secret residue cleaning algorithms and custom allocators for Intel SGX using **Rust**.
- Formally verified execution models in **Coq** to ensure no secret leakage to unauthorized parties.

PUBLICATIONS

- **Haobin Hiroki Chen**, W. Huang, J. Lang, H. Chen, M. Vij, M. Marlara, C. Wang, X. Wang. A Hardware-Software Co-Design Intra-CVM Isolation for Cloud Serverless Computing. *Manuscript available upon request*.
- Y. Guo, H. Chen, **Haobin Hiroki Chen**, Y. Luo, X. Wang, C. Wang. BOLT: Bandwidth-Optimized Lightning-Fast Oblivious Map powered by Secure HBM Accelerators. In *ACM CCS (CCS'25)*, 2025.
- **Haobin Hiroki Chen**, H. Chen, M. Sun, C. Wang, X. Wang. PICACHV: Formally Verified Data Use Policy Enforcement for Secure Data Analytics. In *USENIX Security (Sec'25)*, 2025.
- **Haobin Chen**, Y. Yang, S. Lv. Revisiting frequency-smoothing encryption: new security definitions and efficient construction. *Cybersecurity* (7), 15 (2024).
- H. Chen, **Haobin Hiroki Chen**, M. Sun, K. Li, Z. Chen, X. Wang. A Verified Confidential Computing as a Service Framework. In *USENIX Security (Sec'23)*, 2023.

TECHNICAL SKILLS

Programming	Rust, C/C++, Python, OCaml, Haskell
Verification	Verus, Coq/Rocq, Lean4
Systems	TEEs (SGX, TDX, SEV), Linux Kernel, WebAssembly, Confidential Computing
Tools	LLVM, Git, Docker, L ^A T _E X

HONORS AND AWARDS

- 2026 **Research Funds from the Confidential Computing Consortium**
- 2024 **ACM CCS Distinguished Artifact Reviewer**
- 2023 **Nankai Distinguished Bachelor Thesis Award (Top 3%)**
- 2021 **3rd Prize, National College Student Information Security Contest (Top 8%)**
- 2021/22 **Nankai Academically Excellent Student Scholarship (Top 5%)**

ACADEMIC SERVICES

Program Committee (Artifact Evaluation): ACM CCS '24, USENIX Security '25, NDSS '26
External Reviewer: IEEE TIFS, ACM SIGMOD '26, PeerJ Computer Science